Dubhe Summit (ATH-996)

Institutional Alternative Investment Strategy — Quantitative Performance Analysis

Track Record: Mar 2023 – Jul 2026 (41 Months / 3.4 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Alpha (Ann.)	Beta & Corr.
23.25%	**5.16%**	**4.11**	**-4.94%**	**18.32%**	**0.14 | +0.34**
S&P 500: 22.02%	S&P 500: 12.61%	SPX: 1.65 | Sortino: 7.94	SPX: -8.25% | Calmar: 4.71	t-stat: 6.20	Beta t-stat: 2.25

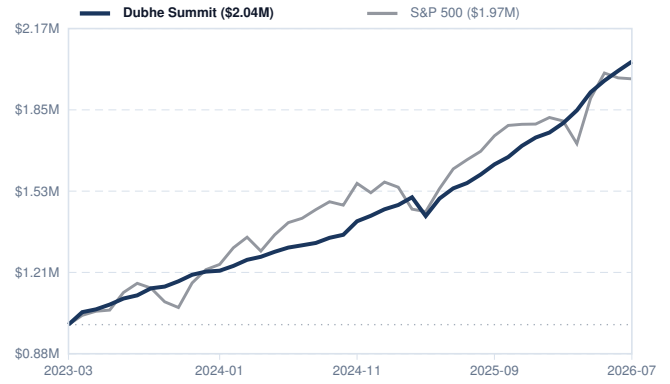
Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Annualized)	**23.25%**	22.02%
Annualized Mean Return	21.21%	20.83%
Annualized Volatility	5.16%	12.61%
Proxy Sharpe Ratio ($r_f = 0\%$)	**4.11**	1.65
Annualized Sortino Ratio	7.94	3.62
Maximum Drawdown (MDD)	**-4.94%**	-8.25%
Calmar Ratio (CAGR / |MDD|)	4.71	2.67
Cumulative Total Return	104.24%	97.41%
Best / Worst Monthly	+4.93% / -4.94%	+10.49% / -5.63%
Positive Months (Hit Rate)	97.6% (40/41)	68.3% (28/41)
Crisis Hit Rate (Fund > 0 | SPX < 0)	**92.3% (12/13)**	–
Up-Mkt Loss (Fund < 0 | SPX > 0)	0.0% (0/28)	–

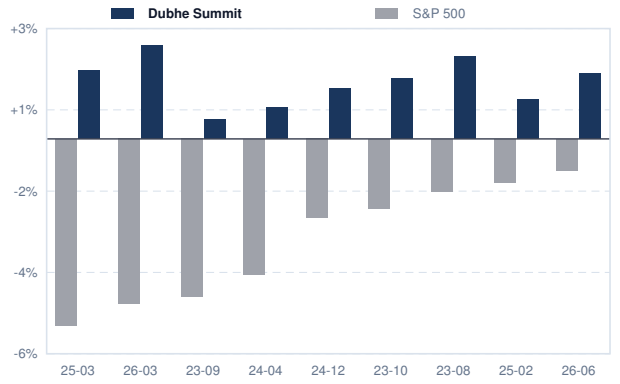
Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**18.32% (t=6.20)**
Market Beta (β_{SPX}) & t-stat	0.14 (t=2.25) ($R^2 = 11.5\%$)
Pearson Correlation (r) to S&P 500	**+0.34** (95% Blk CI: [0.16, 0.54])
Spearman Rank Correlation (ρ)	+0.34 (95% Blk CI: [0.07, 0.59])
Upside / Downside Capture Ratio	Up: **0.56** | Down: **-0.51**
Cond. Up-Market (SPX > 0)	$\alpha = 14.7\%,\ \beta = 0.23$
Cond. Down-Market (SPX < 0)	$\alpha = 6.9\%,\ \beta = -0.26$
Tail Corr. (SPX $< Q_1$, Worst 25%)	$r = -0.10,\ \rho = -0.25\ (n = 10)$
Tail Corr. (SPX $< D_1$, Worst 10%)	$r = -0.56,\ \rho = -0.60\ (n = 4)$
Worst 20% SPX Months Avg. Return	Fund: +1.72% vs SPX: -3.09%
Cross-Asset Beta (WTI / BTC)	WTI β: 0.03 | BTC β: 0.01
Ljung-Box Serial Correlation	$p_1 = 0.15,\ p_3 = 0.53$ (Pass)

Cumulative Growth of $1M USD (Fund vs. S&P 500)



Worst 20% S&P 500 Months



Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2023	-	-	4.9%	1.1%	1.8%	2.2%	1.2%	2.5%	0.6%	1.8%	2.2%	1.1%	**21.0%**
2024	0.3%	1.6%	2.0%	0.9%	1.6%	1.3%	0.7%	0.7%	1.5%	0.9%	4.0%	1.5%	**18.3%**
2025	1.8%	1.2%	2.1%	-4.9%	4.8%	2.7%	1.4%	2.2%	2.5%	1.8%	2.6%	1.9%	**21.7%**
2026	1.1%	2.1%	2.8%	3.9%	2.4%	2.0%	1.8%	-	-	-	-	-	**17.3%**

Fund Terms & Commercial Overview

Vehicle:	Commingled Fund	Return Basis:	Net of Fees	Fee Structure:	2% Mgmt / 20% Perf	Liquidity:	Quarter Liquidity
Lock-Up Period:	No Lockup	Min. Investment:	1M USD	Current AUM:	250M	Domicile:	Cayman Islands

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Risk Disclosures, Legal Notice & Quantitative Appendix

Fund Reference: Dubhe Summit (ATH-996) (Mar 2023 - Jul 2026 (41 Months / 3.4 Years))

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Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [0.11, 0.57] | Blk: [0.16, 0.54]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, IID vs. Block $B = 3$)	Std: [0.01, 0.60] | Blk: [0.07, 0.59]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (PSD Projection)	$\rho_{\text{wc}} = 0.54$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.15,\ p_3 = 0.53$ (Pass)
Conditional Up-Market OLS	Annualized Alpha & Beta estimated during positive S&P 500 months ($n = 28$)	$\alpha = 14.7\%,\ \beta = 0.23\ (t_\alpha = 3.54)$
Conditional Down-Market OLS	Annualized Alpha & Beta estimated during negative S&P 500 months ($n = 13$)	$\alpha = 6.9\%,\ \beta = -0.26\ (t_\alpha = 0.64)$
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = 0.03\ (t = 1.39),\ \alpha = 20.9\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.01\ (t = 0.77),\ \alpha = 20.7\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as $\text{CAGR} = \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as $\text{MDD} = \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

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